A Leakage-Aware Multimodal Crop-Storage Modeling Pipeline

Simulation Audit and Public Tomato Scenario Study

UROP Project Report

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1. Introduction

Crop quality can change during storage as temperature, relative humidity, crop condition, and volatile compounds change. This project explores a monitoring workflow that combines AeroSense environmental readings with responses from a nine-channel MQ sensor array (MQ2, MQ3, MQ4, MQ5, MQ6, MQ7, MQ8, MQ9, and MQ135). The intended outputs are a current quality state and, when inspection histories are available, risk over a future inspection interval.

The present five-crop dataset covers wheat, maize, table potato, cured dry onion, and spinach. It contains 120,000 rows, but every reading, timestamp, lot, inspection label, and failure outcome in that dataset is simulated. Its 500 simulated starting lots branch into six storage paths, each with 40 repeated observations. Thus, the rows are not 120,000 independent crop samples.

The project has two aims: build a reproducible sensor-to-model pipeline and test its evaluation safeguards. Results below describe reconstruction of the simulator only. They are not validated crop accuracy, food-safety decisions, or evidence that a storage intervention adds a known number of days.

2. Literature Survey

The reviewed sources provide useful measurement examples, but none supplies the complete paired dataset required for real crop-quality and remaining-life validation.

Study / resource	Measured data and target	Relevance and limitation
Tomato e-nose dataset, Zenodo 15469472 [1]	12 Rs/R0 gas-response channels, weight, colour, storage day and scenario.	Supports a separate tomato storage-scenario experiment. Scenario/day is not an inspected spoiled label; AeroSense temperature/RH is absent.
Nine-MQ food recordings, Zenodo 17285312 [2]	Short recordings with the nine requested MQ channels; tomato and banana day-coded sessions.	Useful for channel-scale exploration. No verified lot key, matched environment history, or independent spoilage inspection.
Potato storage archive, Mendeley [3]	Temperature, RH, tuber-core temperature and CO2 histories across nodes/modes.	Useful environmental context. MQ and quality summaries are not joined to each sensor row by verified lot/time.
UC Davis postharvest fact sheets [4–6]	Crop-stage and intended-use storage guidance for tomato, potato and banana.	Informs plausible scenario settings; it does not provide training labels or universal spoilage cutoffs.

Research gap

MQ modules are cross-sensitive response devices. Raw ADC-like readings or Rs/R0 values are not automatically gas concentrations, and their response depends on calibration and environmental conditions. The public studies are therefore kept separate. No unrelated tomato, potato, or banana rows are joined to create a fictional measured sample.

3. System Model

3.1 Data and prediction setup

The main experiment loads the supplied five-crop simulated CSV. Predictors are divided into environment-only, MQ-only, and combined groups. At a prediction time, the pipeline uses current and documented through-current history fields. It excludes status and inspection defects, failure intervals, event/follow-up/remaining-life targets, post-failure flags, lot/path IDs, scenario names, fictional absolute dates, and future readings. Grain-moisture fields are missing for vegetable rows and are handled within the fitted pipeline.

3.2 Validation and models

Starting lots, not individual rows, are the split unit. The fixed holdout contains 400 training lots and 100 test lots; every path and time point from one starting lot stays together. Five-fold grouped validation tunes the combined Random Forest and XGBoost using training lots only. A majority-class model is the reference baseline. The comparison reports macro F1, spoiled-class recall, log loss and multiclass Brier score. The starting-inspection feature was assessed separately because it must be collected at deployment if used.

3.3 Time-to-event model

The discrete-time hazard model predicts failure at the next recorded inspection interval from features available at that interval's start. A first simulated spoiled inspection is an event within the recorded inspection interval. Trajectories still acceptable at monitoring end are right-censored; they are not given an invented failure date. Post-failure readings are excluded. XGBoost AFT is retained as a comparator. Future-path projections require explicit future environmental and all nine future MQ values; current MQ readings are never silently copied forward.

3.4 Runnable demo

The local browser interface loads saved models and accepts manual or CSV inputs. Start it from the project folder with: .\.venv\Scripts\python.exe -m crop_prototype.web. The command-line simulated example is: .\.venv\Scripts\python.exe -m crop_prototype.cli predict --input examples\simulated_prediction_input.json. Both are labelled SIMULATION ONLY; neither connects directly to live hardware.

3.5 Additional three-crop simulation

A separate Edition 2 V3 exercise adds 100 generated trajectories and eight readings per trajectory for tomato, table potato, and banana. Tomato/banana MQ recording medians anchor response scale only; potato MQ is generated from a generic scale prior. All environments and labels are synthetic. V3 uses acceptable/warning/high-risk simulator states and its own grouped 80/20 sample split. It is separate from the five-crop dataset and does not create extra measured lots.

4. Results and Discussion

All figures below are held-out simulator results. The test set contains repeated observations from 100 starting lots, not 24,000 independent samples.

Inputs / model	Macro F1	Spoiled recall	Log loss	Brier
Majority baseline	0.1794	0.0000	1.0962	0.6650
Environment / XGBoost	0.8360	0.8445	0.3921	0.2382
Nine MQ / XGBoost	0.9251	0.9526	0.1738	0.1084
Combined / Random Forest	0.9309	0.9587	0.1783	0.1055
Combined / XGBoost	0.9300	0.9562	0.1622	0.1004

The combined XGBoost has slightly better probability scores (log loss and Brier), while Random Forest has marginally higher macro F1 and spoiled recall. Whole-lot paired bootstrap intervals for the differences in macro F1 and spoiled recall include zero; those ranking differences are uncertain. XGBoost is a reasonable probability-focused demo choice, with Random Forest retained as a comparable benchmark.

Crop	XGBoost macro F1	Spoiled recall
Wheat	0.9185	0.9536
Maize	0.9232	0.9394
Table potato	0.9400	0.9621
Cured dry onion	0.9165	0.9391
Spinach	0.9506	0.9718

Next-inspection risk and separate real-tomato task

On the simulated held-out intervals, XGBoost hazard Brier score was 0.01545 and interval AUC was 0.98036; discrete logistic Brier was 0.05504 and AUC was 0.98297. These assess next-check classification, not days-left accuracy. The separate real tomato dataset predicts its recorded storage scenario, not spoilage: leave-one-day-out XGBoost macro F1 was 0.498; on nine separately tracked tomatoes, XGBoost macro F1 was 0.278. This small transfer check is weak and is not a quality validation.

Edition 2 V3 held-out simulation comparison

Crop	Model	Macro F1	High-risk recall
Tomato	Random Forest	0.787	0.630
Tomato	XGBoost	0.746	0.667
Table potato	Random Forest	0.614	0.400
Table potato	XGBoost	0.580	0.320
Banana	Random Forest	0.687	0.633
Banana	XGBoost	0.745	0.733

V3 uses 20 held-out sample trajectories per crop. Whole-dataset simulated counts (acceptable / warning / high-risk) are tomato 393/271/136, potato 380/286/134, and banana 368/287/145. No classifier wins for every crop, and potato high-risk recall is low. These scores measure this generator only; V3 labels are not independent inspections.

5. Concluding Remarks

The project has an end-to-end, runnable demonstration for data inspection, grouped evaluation, current-state probabilities, and interval-based risk. The lot-grouped split, leakage exclusions, censoring logic, explicit-input interface, and V3 sample-level grouping are checked by 30 focused tests. The simulator results show that the pipeline can learn its generated patterns. They do not establish that its labels, sensor responses, or accuracy transfer to crops in storage.

6. Future Work

The next research milestone is a prospective, crop-specific study. Collect temperature and RH plus all nine MQ responses from the same identified lot at matched times; record device, calibration, warm-up, and measurement window. Independent assessors should apply a written crop-specific rubric at scheduled inspections: acceptable means sound/marketable without active decay; warning means early deterioration or limited defects needing recheck; high-risk means severe decay or nonmarketable condition. Record the crop-specific evidence and assessor/time. Store last acceptable and first unacceptable inspection as an interval; if no failure occurs, record monitoring end as censoring. Hold out entire lots and preferably a later harvest or device. First test on real-only training; add simulation only if it improves a separate untouched real-lot test.

A temperature/RH what-if is only a conditional model projection when explicit future MQ values are also supplied. It does not prove that cooling causes longer shelf life. Real paired measurements and independent inspections are required before such a claim.

References

[1] Zenodo, Tomato e-nose and laboratory data, record 15469472. https://zenodo.org/records/15469472

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[7] XGBoost Developers, AFT Survival Analysis. https://xgboost.readthedocs.io/en/stable/tutorials/aft_survival_analysis.html